FTM Theodolite Survey and Calibration Procedures

Surveying the FTM Theodolite Site

- Obtain authorization from the airport manager or control tower to access the runway for 30 minutes.
- Ensure the theodolite battery is fully charged.
- Place the theodolite at the runway centerline point and threshold.
- Level the theodolite and power it on.
- Configure the EDM device on the theodolite and verify measurement units in feet.
- Align the theodolite lens with the runway centerline and set it as zero reference.
- Use reference tables to calculate distance to prism pole.
- Second technician walks along runway centerline to calculated distance, positions prism pole.
- Mark location for calibration use (ensures lens aligned with glide path).
- For offset approach: rotate theodolite 90° and place prism pole accordingly, then mark location.

TLS Calibration Procedures (from Calibration Manual)

The calibration process includes system setup, pre-flight calibration, pre-flight planning, calibration profiles, file generation, validation, and quality checks. Two technicians are required. Proper coordination with airport authorities is mandatory before accessing the runway.

- Verify system power and configuration (Rack A active, cables connected, theodolite powered and leveled).
- Run Flight Test Monitor (FTM) software from Maintenance Interface (MI).
- Ping theodolite and verify elevation/horizontal data displayed.
- Pre-flight planning with pilot/operator: brief profiles and safety procedures.
- Flight profiles: constant altitude runs and glide path descents at 3.0°.
- Track aircraft with theodolite, record data with FTM software.
- Generate calibration file (.cal.dat) using SysCal tool.
- Validate calibration with Look tool (check within tolerances).
- Perform alternative calibration if visual tracking is not possible.
- Conduct post-calibration quality checks and document results.

Illustrations

Figure 1: Theodolite positioned at runway centerline and threshold.

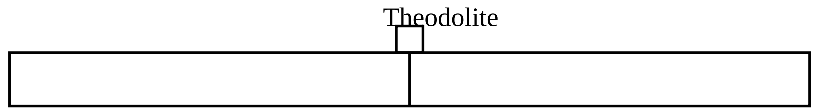


Figure 2: Placement of prism pole along the centerline.

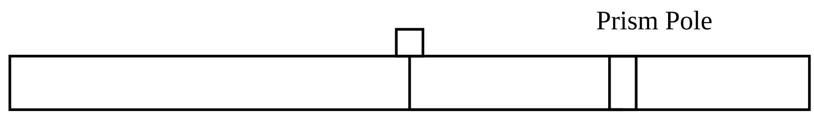
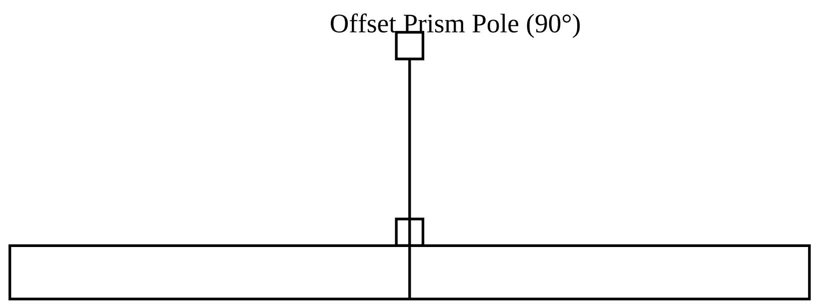


Figure 3: Offset approach with prism pole at 90° to the runway centerline.



End of Document